RUSSIAN-AFRICAN FORUM OF YOUNG SCIENTISTS
"FUTURE ENGINEERS OF THE WORLD – THE FOUNDATION OF SUSTAINABLE DEVELOPMENT"

FAIR MINING DEAL

A constructor of balanced Russian–African investment agreements in subsoil use

Track 2. "RAW MATERIALS DIPLOMACY": Economics, Law and Partnership for Sustainable Development

Peter the Great St. Petersburg Polytechnic University (Russia) · University of Lagos (Nigeria)

Saint Petersburg
2026

RUSSIAN-AFRICAN RAW MATERIALS DIALOGUE

The team

A bilateral index needs two panels — so the team itself is bilateral

1

Sofia A. Afanasieva Team leader · Methodology and development

Peter the Great St. Petersburg Polytechnic University, Russia

Architecture of the clause checklist and of the index calculation core; project management.

2

Polina S. Antonyuk Economist · Analyst

Peter the Great St. Petersburg Polytechnic University, Russia

Derivation of the weight vectors from the documentary corpora; term-sheet economics and market.

3

Esabu Blessing Regional specialist · African expert panel

University of Lagos, Nigeria

African mining law and local context; coding and validation of the host-state corpus.

Why this team

The index is bilateral by construction: the same agreement is scored twice, once against what African states legislate and once against what investors report as risk.

A mixed Russia–Nigeria team reproduces that structure inside the project.

POLYTECH
Peter the Great St. Petersburg Polytechnic University

UNIVERSITY OF LAGOS
IN DEED AND IN TRUTH

Joint abstract submitted to the Russian–African Forum of Young Scientists, 2026.

Relevance: asymmetric deals are being repriced — at a cost

The problem is not a shortage of mining agreements in Africa. It is that a large share of them is renegotiated, suspended or arbitrated within a decade of signature.

45%

of the 63 new ICSID cases in 2025 came from mining (24%) and oil & gas (21%)

24%

of those new cases named a Sub-Saharan African state — more than any other region

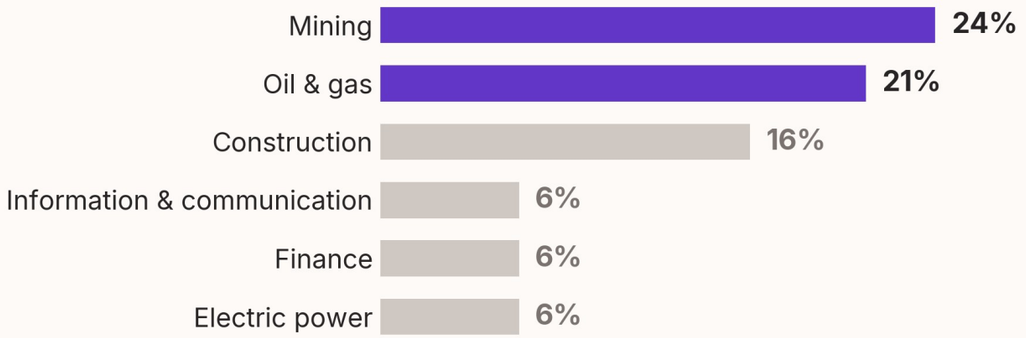
$430M

paid by Barrick to Mali in Nov 2025 to settle the Loulo–Gounkoto dispute

~40%

of the mineral revenue they could collect is what African states actually capture

Sectors of the 63 new ICSID cases registered in 2025



The 2022–2025 repricing wave

2022	**Zimbabwe —** bans the export of raw lithium
2023	**Mali —** new Mining Code: state share raised to 35%, applied immediately
2023	**Namibia —** bans the export of unprocessed critical minerals
2024	**DR Congo —** state stake in a cobalt–copper JV renegotiated from 32% to 70%
2025	**Botswana —** De Beers state share 25% → 30%, rising to 40% within five years
2025	**Gabon —** prohibits manganese ore exports without value addition
2025	**Ghana —** maximum mining lease cut from 30 to 15 years

Sources: ICSID Caseload Statistics 2025 (World Bank); UNECA, Africa's critical mineral resources; GIS Reports, Africa's new resource nationalism (2025); Afronomicslaw, Barrick v. Mali (ICSID ARB/25/2).

Why now: Russia and African states are signing a new generation of subsoil agreements. Terms fixed today without a symmetry check become the disputes of 2030.

RUSSIAN-AFRICAN RAW MATERIALS DIALOGUE

What already exists — and what is missing

Parameter	FAIR MINING DEAL (we)	EITI	Resource Governance Index (NRGI)	Fraser Institute Index	IMF FARI model
Unit of analysis	**A clause of one specific draft agreement**	Country-level payment flows	Country governance systems	Jurisdiction attractiveness	Fiscal regime cash flows
Whose viewpoint	**Both at once: host state and investor**	Neutral disclosure	Host state	Investor companies	Government
When it is applied	**Ex ante — before signature**	Ex post, annual reports	Ex post, 18 countries (2021)	Ex post, annual survey	Ex ante, fiscal terms only
Output	**Symmetry gap Δ + a list of term-sheet edits**	Disclosure reports	Score and ranking	Ranking of jurisdictions	Revenue and NPV projections
Reproducible without expert discretion	**Yes — 3-level checklist, weights from public documents**	n/a	Partly — expert survey	No — opinion survey	No — model plus confidential data
Covers non-fiscal fairness: technology transfer, local content, rehabilitation	**Yes — 7 weighted blocks**	No	Partly	No	No

Conclusion: no existing instrument scores a specific draft agreement from both sides before it is signed. Fair Mining Deal fills that gap — and returns concrete clause edits instead of a ranking.

Sources: eiti.org (55 implementing countries); NRGI Resource Governance Index 2021; Fraser Institute Annual Survey of Mining Companies 2025; IMF, Fiscal Analysis of Resource Industries.

Concept: fairness measured as symmetry

Novelty — the index does not ask “is this deal good?”. It asks “do the two sides see the same deal?”, and it asks the question of the contract text, not of an expert.

1

Draft agreement

The document already on the negotiating table — the only input required.

2

Clause checklist

Seven blocks of verifiable provisions, each scored on three levels.

3

Two weight vectors

What each side counts as important — read off two published, countable instruments.

4

Two indices

The same deal scored from the host state’s side and from the investor’s side.

5

Verdict and edits

The gap between them becomes a revision-risk class and a short list of edits.

KEY INNOVATION

Fairness is not one number.

It is the distance between two numbers.

A single “fairness score” always belongs to somebody. Two scores of the same text, and the gap between them, belong to the negotiation.

What the negotiating team receives

- a revision-risk class for the agreement as tabled;
- the blocks where the two sides diverge most;
- a ranked list of clause edits that close the gap at the lowest cost to the winning side;
- a record of the assessment that can be repeated by the other party — the same text gives the same score.

The checklist deliberately avoids formulas at the negotiating table: every input is a provision a lawyer can point to in the draft.

Technical implementation

Weights counted off two published instruments; the index applied to a real agreement whose outcome is known.

$$FMDI_k = \Sigma\,(\,w_i^k \times S_i^k\,),\quad i = 1...7 \qquad \Delta = FMDI_{host} - FMDI_{inv}$$

$S_i^k \in [0; 100]$ — block score from the clause checklist: **0** absent · **50** declarative · **100** measurable obligation with a deadline and a sanction. Two officials reading the same draft return the same score.

Host-state priority

what states actually legislate

All 31 mining-sector measures adopted by African states between 2022 and the first half of 2026 that we were able to document, each coded to the one block it acts on. The weight is that block's share of the side's own agenda.

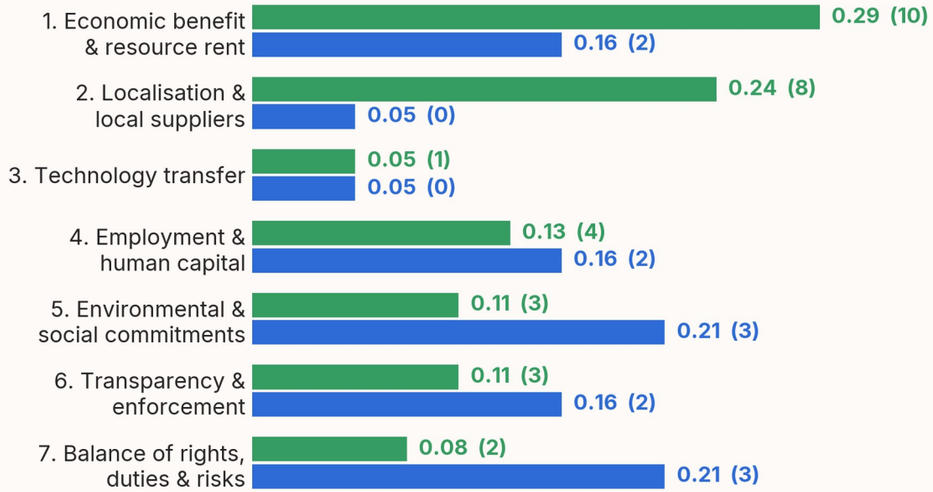
Cross-check: the block states legislate most (rent, 10 of 31) and the block investors defend most (legal stability) are exactly the two that collided in Mali in 2023–2025.

Investor priority

what the industry's own risk instrument asks about

The 15 policy factors of the Fraser Institute Annual Survey of Mining Companies 2025 (256 respondents, 68 jurisdictions), each mapped to the block it concerns; 3 factors are not contract provisions and are recorded out of scope.

Neither localisation nor technology transfer appears among the 15 factors at all — that absence is itself the asymmetry the index measures.

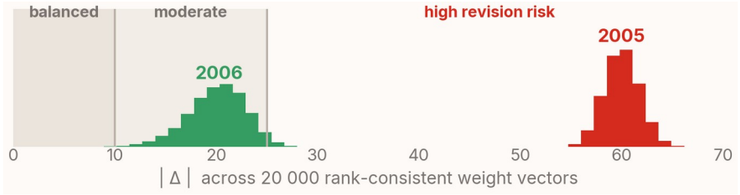


RETROSPECTIVE VALIDATION · Liberia — Mittal Steel MDA and its 2006 amendment · 24 positions read from the texts

	FMDI host	FMDI investor	|Δ| → verdict
As signed, 17 Aug 2005	26.9	86.8	60.0 → **HIGH revision risk**
Renegotiated, 28 Dec 2006	52.1	74.4	22.3 → **moderate asymmetry**

The 2005 text also fails the level rule (host 26.9 < 40). It was reviewed within a year and renegotiated in four rounds — the index flags it before that happened.

ROBUSTNESS TO THE WEIGHTING



2005 stays in "high risk" in **100%** of draws, 2006 in "moderate" in **97%**.

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Contribution to sustainable development

The project does not declare compliance with the SDGs — it turns three of them into scored, sanctioned clauses of a contract.

8

DECENT WORK AND ECONOMIC GROWTH

Blocks 2 and 4 score local employment, supplier quotas and training as obligations with figures, deadlines and penalties — not as goodwill paragraphs.

9

INDUSTRY, INNOVATION AND INFRASTRUCTURE

Block 3 scores technology transfer: process documentation, the right to use the technology after the term expires, and local processing capacity.

12

RESPONSIBLE CONSUMPTION AND PRODUCTION

Block 5 scores environmental and rehabilitation commitments by whether they are funded and enforceable, not by whether they are promised.

EFFECT OF A SYMMETRY CHECK BEFORE SIGNATURE

Economic impact

Extractive industries produced 45% of all new ICSID cases in 2025. A deal that survives without renegotiation is cheaper for both sides than one that is arbitrated: the Barrick–Mali settlement alone cost $430M before any award was made.

Social impact

Employment, training and local-supplier commitments only score full marks when they carry a number, a deadline and a sanction. The index rewards exactly the wording that local communities can later enforce.

Environmental impact

Rehabilitation and environmental obligations are scored by whether money is set aside and by whom, so closure costs are priced into the agreement instead of being left to the host state.

Without compromising the future: the instrument adds nothing to the cost of extraction. It only moves obligations that would otherwise be negotiated after a conflict into the text that is signed before it.

Target audience and market

The customer is a negotiating team that has to sign, approve or ratify a subsoil agreement — usually within 3–9 months.

Host-state side

- ministries of mines and natural resources
- national investment promotion agencies
- state mining companies and sovereign funds
- parliamentary committees ratifying conventions

Investor side

- Russian mining, oil & gas and exploration companies
- their legal, tax and financial advisors
- insurers and lenders pricing political risk
- joint-venture partners entering an existing licence

Third parties

- development banks and export credit agencies
- arbitration and consulting practices
- EITI multi-stakeholder groups and civil society
- universities teaching mining law and economics

SCALE OF THE PROBLEM

54

African states; 55 countries worldwide implement the EITI standard

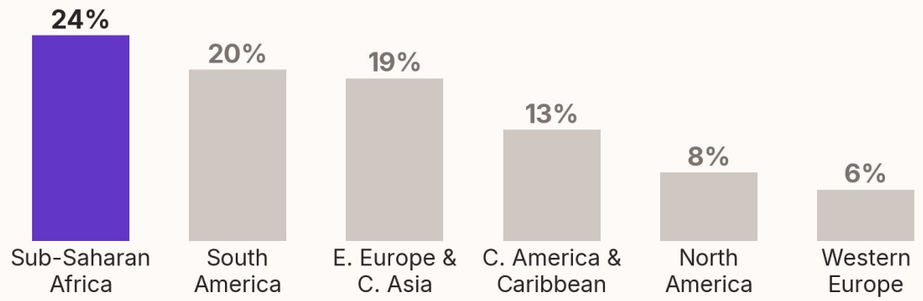
1 085

ICSID arbitration and conciliation cases registered since 1972

~30%

of world mineral reserves are in Africa: 55% of cobalt, 48% of manganese

Where new 2025 ICSID cases were filed, by respondent state



Portrait of the primary customer. A working group of 5–15 people — a lawyer, an economist and a geologist on each side — with a draft convention, a deadline and no reproducible way to answer the one question their minister or board will ask: “is this balanced, and where exactly is it not?”

Delivery models: annual licence to a state agency or company · per-deal advisory assessment · free academic version for universities.

Sources: ICSID Caseload Statistics 2025; eiti.org; UNECA.

Adaptation to the conditions of the region

The instrument is not equipment, so it is not the climate that limits it — it is data, law, language and staff capacity.

1 Data and infrastructure

- Input is the contract text alone — no company financial statements, no geological database, no production data the parties will not disclose.
- Offline desktop and spreadsheet version under 5 MB: unstable electricity, low bandwidth and field conditions do not block an assessment.
- Nothing leaves the ministry: an agreement under negotiation is confidential, so no cloud upload is required.

2 Law and language

- The checklist is recalculated against the national mining code — Mali Law 2023-040, Nigeria NMMA 2007, DR Congo Code 2018 — so a clause is scored against what that country already requires.
- English, French and Russian: Francophone West Africa is more than half of the target jurisdictions.
- Currency and banking channel of settlement is itself a scored item, so sanction exposure is assessed rather than ignored.

3 People and institutions

- Three-level scoring needs a lawyer who can read the draft, not a modeller: two different officials reading the same text arrive at the same score.
- Two days of training are enough to run an assessment; the weight vectors are supplied, not derived locally.
- High staff turnover in ministries is survivable — the record of an assessment is reproducible by whoever comes next.

Why would it work there, and not just in the lab?

Because its only input is a document that every negotiation already has on the table. The assessment runs on a laptop in a ministry office, in the language the draft is written in, by the lawyer who is reading it anyway — and it produces the same answer for the other side of the table, which is what makes the result usable as a negotiating document rather than as advice.

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Implementation plan and economics

Three years from method to a licensed instrument — no capital equipment and no field work required.

01 2026–2027

Method and validation

- ✓ **corpus v0 (31 measures), both weight vectors, robustness tests**
- checklist v1.0 — about 70 verifiable provisions across the 7 blocks
- corpus extended to the full UNCTAD record, double-coded with Krippendorff's α
- retrospective validation on 15–20 published contracts; the thresholds are calibrated on that sample

02 2027–2028

Pilot and tooling

- assessment of one real Russian–African draft with a partner ministry or company
- web calculator plus an automatically generated term-sheet memo
- expert panels convened to validate the document-derived weights, not to replace them

03 2028–2030

Scaling

- jurisdiction library for 10 African states, interface in EN / FR / RU
- licences to state agencies, companies and advisors; free academic version
- annual re-weighting as mining codes change

ESTIMATED BUDGET

RUB 6.5–8.0 M over three years

About USD 77–95 k at the rate of 11 September 2026. Coding and double-coding of the corpus, legal translation into French, web development, validation data and one validation panel. No capital equipment and no field work.

POTENTIAL PAYBACK

One avoided renegotiation

Barrick's settlement with Mali was USD 430 M, on top of a suspended mine, seized gold and an ICSID case running from January to November 2025. The instrument costs four orders of magnitude less.

RESOURCES AND PARTNERS NEEDED

A jurisdiction, a draft and two panels

SPbPU and University of Lagos as research base; a ministry of mines or investment agency for the pilot; a Russian subsoil user; an arbitration practice; an EITI national secretariat; a research supervisor.

Each stage delivers a usable artefact: checklist v1.0 is already a negotiating aid before the weights are final, and the retrospective validation is publishable on its own.

Conclusion and invitation to dialogue

Fair Mining Deal turns “is this agreement fair?” into a reproducible number computed from the clauses of the contract itself — from both sides at once — and converts the gap between the two answers into a short list of edits.

Renegotiation risk is then priced before signature, not litigated after it.

WHAT THE TEAM NEEDS RIGHT NOW

- a pilot jurisdiction and access to one real draft agreement, under NDA;
- 10–12 experts per panel, Russian and African, to validate the derived weights;
- a partner ministry, investment agency or subsoil-user company;
- a research supervisor in mining law or resource economics;
- introductions to an arbitration practice and an EITI national secretariat.

TEAM CAPTAIN

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We are ready to answer your questions.

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Additional materials

Not shown in the main speech — held in reserve for the jury's questions

Weighting protocol

- One measure — one block: a measure is coded to the block that is the direct object of the rule, never to a secondary effect.
- The weight of a block is its share of that side's own agenda, Laplace-smoothed (+1 per block) so that no block leaves the index.
- No one is asked what matters: both instruments are public and enumerable, so a third party recounts them and obtains the same vector.
- Expert panels enter at stage 2 to validate the derived weights, not to produce them.

The corpus at a glance

block	state	inv.
1. Rent and benefit	10	2
2. Localisation	8	0
3. Technology transfer	1	0
4. Employment	4	2
5. Environment & social	3	3
6. Transparency	3	2
7. Rights and risks	2	3

Σ 31 measures, 2022–H1 2026
Σ 12 of 15 survey factors in scope

What the verdict does and does not depend on

Δ for the demonstration case under four weighting rules

Frequency (base case)	-28.6
Rank-order centroid	-28.0
Rank sum	-28.2
Equal weights	-30.7

Weights. 20 000 vectors drawn uniformly from the simplex subject only to the observed rank order: median |Δ| 27.8, 90% interval 23.5–32.8; 86% inside "high revision risk".

Coding. Re-assigning the 15 arguable measures at random: |Δ| stays 25.7–29.6, 100% in the same class.

Clause scores. Free the scores too, keeping only the documented pattern: the host index stays below the investor index in 99.9% of cases and outside "balanced" in 88% — but the high-risk class holds in only 8%. The class belongs to this draft; the direction belongs to the method.

Interpretation — and the status of the thresholds

Symmetry: |Δ| ≤ 10 balanced partnership; 10–25 moderate asymmetry; above 25 high probability of revision. **Level:** if the lower of the two indices falls below 40, the terms are unacceptable for that side.

The three thresholds are provisional and are not derived. Stage 1 calibrates them on the retrospective sample: the cut-offs that best separate agreements later renegotiated, suspended or arbitrated from those that ran to term. Until then Δ is reported as a continuous value and the per-block gaps carry the message.

Known limitations

- The two instruments count different things — rules adopted on one side, survey factors on the other. Weighting the factors by their published deterrence shares is the next step.
- The fine ordering of the investor vector is unstable: re-coding leaves its top pair unchanged in only 53% of draws (12 in-scope items against 31). It does not move the verdict, which is driven by the state-side spread.
- One weight vector for the whole continent; per-jurisdiction vectors need a per-jurisdiction corpus, and the corpus rests on English-language sources.

Method. Rank-order centroid and rank-sum weights follow Barron F. H., Barrett B. E. // Management Science. 1996. Vol. 42(11). P. 1515–1523. The 11 references of the study are listed in the submitted abstract. The full coded corpus, the checklist positions and the robustness tables are in the weighting appendix.

Instruments and data sources. UNCTAD Investment Policy Monitor · Fraser Institute Annual Survey of Mining Companies 2025 · ICSID Caseload Statistics 2025 · UNECA · GIS Reports · Afronomicslaw, Barrick Mining v. Republic of Mali (ICSID ARB/25/2) · Hogan Lovells, Freshfields and Pinsent Masons Africa mining reform trackers · eiti.org · NRGI Resource Governance Index 2021.